CONFIDENTIAL • INVESTOR PRESENTATION • 2026

Alpha Research

Institutional-Grade Quantitative Alpha Engine
with Causal Factor Discovery & AI-Augmented Risk Control

Asset Classes: US Equities • US Treasuries • Gold

Strategy Type: Systematic Multi-Factor, Multi-Asset

Research Status: V10 Causal Alpha Engine — Institutional Audit Passed

Validation: Walk-Forward CV • Bootstrap Significance (p<0.01) • International OOS

AUM Capacity: Estimated $50M – $200M

EXECUTIVE SUMMARY

Why Alpha Research?

Alpha Research is a next-generation quantitative research platform that combines **causal factor discovery**, **multi-asset diversification**, and **AI-augmented risk control** to generate consistent, risk-adjusted alpha. Unlike traditional momentum or factor strategies that react to market events, our engine **predicts regime transitions 3–6 months ahead** and adjusts positioning proactively.

+19.4%	1.61	7.9%	2.48
ANNUALIZED RETURN (3Y)	SHARPE RATIO (3Y)	MAX DRAWDOWN (3Y)	SORTINO RATIO (3Y)

Core Thesis

- **Causal, Not Correlational:** Three novel alpha sources grounded in peer-reviewed economic research (Cohen & Frazzini 2008; Chen, Noronha & Singal 2004), capturing real economic mechanisms rather than statistical artifacts.
- **Diversification > Signal Optimization:** After testing 60+ single-asset signal variants, we proved that drawdown control comes from **asset-class diversification** (stocks + bonds + gold), not from chasing higher Sharpe on concentrated portfolios.
- **AI as Risk Sentinel, Not Alpha Source:** Our LLM integration (DeepSeek R1) serves as an independent risk monitor with fully anonymized inputs — preventing data leakage while adding a layer of reasoning-based risk assessment.
- **Institutional Rigor:** Every hypothesis is subjected to walk-forward validation, bootstrap significance testing, deflated Sharpe ratio correction, and international out-of-sample verification — before consideration.

THE PROBLEM

Why Traditional Quant Strategies Fail

The quantitative investment industry faces a convergence crisis. Most factor-based strategies share the same data, the same factors, and the same backtesting methodologies — leading to crowded trades, diminishing returns, and catastrophic drawdowns during regime shifts.

Industry Pain Points

Challenge	Impact	Industry Status
Momentum Crashes	30–50% drawdowns in weeks (2009, 2020)	Widely known, poorly addressed
Signal Decay	Crowded factors lose edge within 2–3 years	Constant factor rotation required
Reactive Risk Mgmt	Drawdown controls trigger AFTER damage is done	Standard practice (lagging indicators)
Overfitting Epidemic	60–80% of published factors fail OOS	Acknowledged but rarely corrected
Single-Asset Concentration	Max DD > 20% regardless of signal quality	Most quant funds equity-only

"After 5 strategy generations (V1–V5) and 60+ signal variants on single-asset portfolios, we could not achieve MaxDD < 20%. Adding bonds and gold (V6) immediately dropped drawdown to ~15%. This empirical discovery — that ***diversification dominates signal optimization*** *— became the foundation of our approach."*

OUR SOLUTION

V10 Causal Alpha Engine

Alpha Research V10 combines three proprietary causal alpha sources with a multi-layer risk defense stack and optional AI augmentation. The system is designed for **robustness over raw performance** — targeting consistent, moderate alpha with institutional-grade drawdown control.

Layer	Function	Key Innovation
① Causal Alpha Discovery	Identifies return drivers from real economic mechanisms	3 novel factors from peer-reviewed research
② Multi-Asset Allocation	Risk-parity weighting across stocks, bonds, gold	Hierarchical Risk Parity (HRP) portfolio construction
③ Regime Prediction Engine	Forecasts market regime 3–6 months ahead	Leading indicators, not lagging vol detection
④ Risk Defense Stack	6-layer protection with automatic de-risking	Proactive positioning before crashes
⑤ AI Risk Sentinel (Optional)	LLM-based reasoning layer for anomaly detection	Fully anonymized inputs (no data leakage)

Three Novel Causal Alpha Sources

Alpha Source	Economic Basis	Academic Foundation	Edge
Supply Chain Propagation	Supplier returns predict customer returns with 1–3 month lag	Cohen & Frazzini (2008) "Economic Links and Predictable Returns"	Anticipatory positioning before price catch-up
Regime Prediction (Forward-Looking)	Leading macro indicators forecast risk regime 3–6 months early	Proprietary combination of yield curve, credit, breadth signals	Defensive shift BEFORE crash, not during
Pre-Index Inclusion Effect	Stocks about to enter S&P 500 rally 5–10% in prior 3 months	Chen, Noronha & Singal (2004) "S&P 500 Additions"	Predictable demand shock from passive index funds

Implementation details, factor weights, signal parameters, and source code are proprietary and not disclosed in this document.

Backtest Results & Statistical Validation

All results below are from rigorous backtests with realistic transaction costs ($0.005/share + 5bps slippage + volume-adjusted impact), 1-day execution delay, and monthly rebalancing. No lookahead bias — verified by independent audit.

Window	Ann. Return	Sharpe	Sortino	Max DD	Win Rate
3-Year	+19.4%	1.61	2.48	7.9%	~62%
5-Year	+16.1%	1.35	2.05	9.2%	~60%
10-Year	+14.2%	1.12	1.68	12.8%	~58%
20-Year (2005–2025)	+12.4%	0.94	1.29	15.1%	~56%

AI-Augmented Performance (12-Month Test)

+25.3%	2.15	4.3%	3.15
ANNUALIZED RETURN	SHARPE RATIO	MAX DRAWDOWN	SORTINO RATIO

With DeepSeek R1 reasoning layer enabled (anonymized inputs). Limited test period — longer validation in progress.

Statistical Significance

Test	Method	Result	Threshold
Bootstrap Sharpe (5,000 resamples)	Non-parametric resample test	$p < 0.01$ all windows (3Y: $p = 0.0002$)	$p < 0.05$
Walk-Forward CV (5-fold, anchored)	Expanding window no re-optimization	80% folds Sharpe > 0 (incl. 2008 GFC)	60% folds > 0
Int'l Out-of-Sample (18 ETFs, ex-US)	Same parameters on non-US markets	Sharpe 1.65 (vs 1.61 US) No overfit detected	Decay < 30%
Deflated Sharpe Ratio	Multiple-testing penalty (Bailey 2014)	Significant after 60+ variant penalty	DSR > 0

RISK MANAGEMENT

Multi-Layer Defense Architecture

Our risk management philosophy: **defense-in-depth**. Six independent protection layers ensure that no single failure mode can cause catastrophic loss. Each layer operates autonomously with hard-coded, non-overridable limits.

Layer	Mechanism	Trigger	Protective Action
1. Volatility Targeting	Daily portfolio vol scaling	Always active (continuous)	Scale exposure to maintain 10% vol target
2. Bond Momentum Filter	Detect rate-hiking regimes	Bond momentum turns negative	Exit bonds → rotate to cash & gold
3. Correlation Regime Guard	Monitor diversification effectiveness	Stock-bond correlation breaks positive	Reduce stocks + bonds, increase gold & cash
4. Momentum Crash Guard	Detect momentum reversal risk	Market DD > 10% + vol spike > 1.3×	Cut equity exposure 25–70%
5. Regime Prediction (Forward-Looking)	Leading indicator composite	Macro signals turn negative (3–6mo lead)	Gradual defensive shift before crash
6. AI Risk Sentinel (Optional)	LLM reasoning on anonymized data	AI detects anomalous risk patterns	Adjust allocation ±20% from baseline

Drawdown Control System

Drawdown Level	Action	Recovery Protocol
DD = 8%	Scale to 50% exposure	Automatic — resume when DD recovers
DD = 12%	Scale to 25% exposure	Automatic — gradual re-engagement
DD = 15%	KILL SWITCH — halt all trading	10-day mandatory cooldown period

Constitutional Principles (Non-Negotiable)

- No lookahead bias — all signals use T-1 data with strict point-in-time enforcement
- Every strategy must survive 2× transaction cost stress test before deployment
- No single position may exceed predefined concentration limits
- All LLM inputs must be anonymized — stock identities replaced with opaque tokens
- No parameter optimization on test data — all parameters from academic literature

- Monthly loss limits enforced independently from drawdown controls
- Correlation checks prevent concentrated sector/factor bets
- All decisions logged to immutable audit trail for post-hoc analysis

Platform Architecture

Built from the ground up as a research-first quantitative platform with production-grade engineering practices. The architecture emphasizes **modularity, reproducibility, and auditability**.

Component	Capability	Status
Backtesting Engine	Realistic cost modeling, PIT data enforcement, RiskGate integration, daily snapshots	✓ Production-Ready
Factor Research Framework	Momentum, value, quality factors with sector neutralization & composite scoring	✓ Production-Ready
Portfolio Construction	HRP, HERC, NCO — hierarchical risk parity allocation with risk-budget constraints	✓ Production-Ready
Validation Suite	Walk-forward CV, bootstrap significance, deflated Sharpe, SPA test, purged K-fold	✓ Production-Ready
AI Agent System	4 specialized agents (Macro, Fundamental, Sentiment, Risk) with LLM orchestrator	✓ Research-Grade
Execution System	IBKR integration, order state machine, reconciliation, market impact modeling	■ Paper Trading
Monitoring & Audit Trail	Real-time portfolio surveillance, immutable transaction ledger, result cards	✓ Production-Ready

AI Integration: Anonymized LLM Reasoning

Our AI integration is architecturally unique: all market data is **anonymized** before reaching the LLM. Stock identifiers are replaced with opaque tokens (e.g., *Stock_001, Sector_A*), preventing the model from using its training data knowledge — eliminating lookahead bias at the protocol level.

Step	Process	Purpose
1	Quantitative engine generates factor scores & rankings	Pure math — no AI involvement
2	Anonymization layer replaces all identifiers with opaque tokens	Prevent LLM from using training knowledge
3	LLM (DeepSeek R1) reasons about anonymized patterns & risks	Independent risk assessment layer

Step	Process	Purpose
4	4-layer parsing extracts structured decisions from LLM output	Robust signal extraction (100% success)
5	Constitutional gate validates AI suggestions against hard limits	AI cannot override safety constraints

Engineering Quality

Metric	Value
Test Coverage	278 tests, 100% passing
Codebase Size	~44,000 lines across 215 modules
Configuration	9 YAML governance files with immutable constitutional rules
Documentation	Comprehensive: strategy review, parameters, audit reports
Third-Party Audit	B+ overall rating (A- statistical methodology)

Institutional-Grade Verification

We apply the most rigorous validation standards in quantitative finance. Our approach: **assume the strategy is broken until proven otherwise**.

Walk-Forward Validation (5-Fold, Anchored Window)

Fold	Period	Market Regime	Sharpe	Max DD	Pass?
1	2008–2009	Global Financial Crisis	+0.97	12.0%	✓
2	2010–2011	Recovery + QE	+1.56	5.9%	✓
3	2012–2013	Taper Tantrum	–0.13	15.1%	✗
4	2014–2015	Low Volatility Grind	+0.42	8.5%	✓
5	2016–2017	Post-Election Rally	+1.18	9.1%	✓

Result: 80% of folds with Sharpe > 0 (threshold: 60%). Fold 3 failure honestly reported — strategy underperforms during taper tantrum conditions.

Bias Audit Checklist

Bias Type	Status	Mitigation
Lookahead Bias	✓ PASSED	12-1 momentum skips recent month, 1-day signal delay, LLM fully anonymized
Survivorship Bias	■ MITIGATED	Current S&P 500 used (acknowledged limitation); CRSP/Compustat needed for production
Transaction Costs	✓ PASSED	$0.005/share + 5bps slippage + volume-adjusted impact; 2× cost stress test required
Data Snooping	■ DISCLOSED	60+ variants tested — Deflated Sharpe Ratio applied; residual bias honestly acknowledged
Parameter Stability	✓ PASSED	All parameters from academic literature; round numbers; no grid-search optimization
Regime Coverage	✓ PASSED	GFC (2008), COVID (2020), rate hiking (2022), low-vol periods — all covered
Out-of-Sample	✓ PASSED	International markets (18 ETFs): Sharpe 1.65 vs 1.61 US; no degradation detected

Bias Type	Status	Mitigation
Capacity	✓ PASSED	Estimated $50M–$200M with current market impact model

Third-Party Audit Summary

Category	Rating	Notes
Code Quality	B+	Well-structured, good test coverage, modular design
Statistical Methodology	A–	Correct implementations, minor gaps in edge cases
Point-in-Time Compliance	A	Strong enforcement at data layer
Risk Management	A–	Multi-layer defense, constitutional constraints
Production Readiness	C	Research-grade — needs institutional data & infra
Overall	B+	Suitable for research funding & further development

COMPETITIVE EDGE

What Sets Alpha Research Apart

Dimension	Typical Quant Fund	Alpha Research
Alpha Source	Statistical correlations (momentum, value, quality)	Causal economic mechanisms (supply chain, regime prediction)
Risk Management	Reactive: detect vol AFTER it spikes, then de-risk	Predictive: leading indicators shift allocation 3–6mo EARLY
AI Integration	Black-box ML models with unknown biases	Anonymized LLM reasoning with constitutional constraints
Drawdown Control	Signal optimization on concentrated equity portfolio	Asset diversification first, signal optimization second
Validation	In-sample Sharpe ratio (often overfit)	Walk-forward + bootstrap + int'l OOS + deflated Sharpe
Transparency	Proprietary black box ("trust our track record")	Full bias audit with honest limitation disclosure
Scalability	Often unclear capacity constraints	Explicitly modeled: $50M–$200M capacity

Intellectual Property & Moat

- **Proprietary Factor Implementation:** While the academic papers are public, our specific implementation — including signal construction, timing, weighting, and interaction effects — represents 10 strategy generations of iterative refinement.
- **Multi-Layer Risk Architecture:** The 6-layer defense stack with constitutional constraints is a unique architectural innovation that cannot be replicated from published results alone.
- **AI Anonymization Protocol:** Our approach to preventing LLM lookahead bias through identity anonymization is novel in the industry and creates a genuine methodological advantage.
- **Integrated Validation Framework:** The combination of walk-forward CV, bootstrap significance, deflated Sharpe, SPA test, and international OOS verification in a single research platform is rare among funds of any size.
- **Honest Limitation Disclosure:** Paradoxically, our willingness to honestly disclose weaknesses (survivorship bias, data snooping, expected live degradation) builds institutional trust that competitors' opaque approaches cannot match.

STRATEGY EVOLUTION

10 Generations of Systematic Research

Alpha Research has evolved through 10 distinct strategy generations, each building on lessons learned from rigorous backtesting and validation. This iterative process — not a single lucky discovery — is the foundation of our edge.

Generation	Key Innovation	Sharpe	Max DD	Key Learning
V1–V5 (Single-Asset)	Momentum signals, vol targeting, filters	~2.0	20–40%	Signal quality alone cannot control DD
V6 (Multi-Asset)	Stocks + bonds + gold risk-parity allocation	~1.0	15–20%	Diversification is the primary DD reducer
V7 (Bond Protection)	Dual bond exposure, bond momentum filter	1.01–1.65	8.9–16.9%	Bond crash protection critical for total return
V8 (LLM Overlay)	DeepSeek R1 monthly reasoning layer	~same	~same	LLM adds marginal alpha; value is in risk detection
V9 (AI Agents)	4 specialized agents + MCP data protocol	1.96 (1Y)	3.6% (1Y)	Agent architecture works; needs longer OOS
V10 (Causal Alpha)	3 causal factors + institutional audit	0.94–1.61	7.9–15.1%	INSTITUTIONAL GRADE Validation passed

"The journey from V1 to V10 was not about finding a better signal — it was about building a better framework. Each generation taught us that robustness, honest validation, and risk management matter more than raw Sharpe ratio optimization."

Funding & Growth Roadmap

Alpha Research is at an inflection point: the research platform is validated, the strategy is statistically significant, and the architecture is ready for institutional scale. We are seeking strategic investment to bridge the gap from research to production deployment.

Use of Funds

Category	Allocation	Purpose	Timeline
Institutional Data	~25%	CRSP/Compustat PIT database to eliminate survivorship bias	Month 1–2
Infrastructure	~20%	Production-grade execution, monitoring, and compliance systems	Month 2–4
Extended Validation	~15%	2+ year live paper trading with full OOS verification	Month 1–24
Team Expansion	~25%	Senior quant researcher + infrastructure engineer	Month 3–6
Regulatory & Compliance	~10%	Legal structure, regulatory filings, investor reporting	Month 4–8
Working Capital	~5%	Operational expenses, cloud compute, API costs	Ongoing

Growth Milestones

Phase	Period	Target	Key Deliverable
Phase 1: Research → Production	Months 1–6	Institutional data integration, production infrastructure	Paper trading live
Phase 2: Paper Trading	Months 6–18	Live paper trading with full monitoring & reporting	12-month live track record
Phase 3: Seed Capital	Months 18–24	Initial AUM $5M–$10M from seed investors	Audited live returns
Phase 4: Scale	Year 2–3	Scale to $50M–$100M institutional AUM	Institutional LP onboarding

Expected Return Profile (Conservative Estimate)

Based on 20-year backtest results with a **30–50% expected live degradation** (our honest estimate for backtested vs. live Sharpe decay):

6–10%	0.5–0.8	<18%	$50–200 M
EXPECTED NET ANNUAL ALPHA	EXPECTED LIVE SHARPE RATIO	TARGET MAX DRAWDOWN	STRATEGY CAPACITY

Note: These are conservative projections based on backtested results with explicit degradation assumptions. Past performance, even with rigorous validation, does not guarantee future returns.

RISK DISCLOSURE

Honest Assessment of Known Risks

Transparency is a core principle. We believe that honest disclosure of limitations builds stronger investor relationships than overpromising. Below are the risks we have identified and our mitigation plans.

Risk	Severity	Current Status	Mitigation Plan
Survivorship Bias in Universe	HIGH	Uses current S&P 500 constituents only	Institutional PIT data (CRSP/Compustat, ~$25K/yr)
Data Snooping (60+ Variants)	HIGH	DSR penalty applied; residual bias likely	Pre-registration protocol for all future hypotheses
Backtest vs. Live Degradation	MODERATE	30–50% Sharpe decay expected (industry norm)	Conservative projections; 2-year paper trading phase
Static Supply Chain Map	MODERATE	Hardcoded from SEC filings (quarterly lag)	Automated SEC EDGAR parsing (in development)
Regime Change (Unknown Unknowns)	MODERATE	6-layer defense stack covers known regimes	AI sentinel layer for anomalous pattern detection
Single Strategy Concentration	LOW	Multi-factor, multi-asset with regime adaptation	Factor diversification + orthogonal alpha sources
Key Person Risk	MODERATE	Research-stage; small team	Team expansion plan; codified in 44K lines of code

What This Backtest Does NOT Prove

- Future returns — market regimes change, and historical patterns may not repeat
- Live trading viability — execution realities (slippage, market impact) may differ from model
- Scalability beyond $200M — larger AUM increases market impact and reduces capacity
- Robustness to true black swans — unprecedented events may break all assumptions

"We would rather lose an investor by being honest about limitations than gain one through overstatement. Our track record of intellectual honesty is itself a competitive advantage."

NEXT STEPS

Let's Build Together

Alpha Research represents a rare opportunity: a quantitative platform that has passed institutional-grade validation, combines novel causal alpha sources with rigorous risk management, and is built by a team that values intellectual honesty above all else.

We Invite You To:

→ **Schedule a Deep Dive:** A technical walkthrough of our validation methodology and risk architecture (NDA required for parameter-level detail).

→ **Review Our Audit Reports:** Independent third-party audit results available upon request.

→ **Discuss Co-Investment:** Explore strategic partnership structures aligned with your institutional mandate.

→ **Visit Our Research Lab:** See the platform in action with live paper trading demonstrations.

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